Sofia Loya

sofia-loya.github.io | ssloya@andrew.cmu.edu | 915–238–0511 | linkedin.com/in/sofia-loya | U.S. Citizen

Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Mechanical Engineering | GPA: 4.0/4.0 | Rales Fellows Program — May 2027
- Selected Coursework: Machine Learning and Artificial Intelligence for Engineers, Introduction to Deep Learning

Texas A&M University — College Station, TX
Bachelor of Science in Mechanical Engineering | GPA: 3.587/4.0 — May 2021

Professional Experience

Medtronic — Mounds View, MN
Quality Engineering Intern | Cardiac Ablation Solutions — *June 2026 – Aug 2026*
- Drove EU Clinical Study compliance by developing seven characterization test methods for a small focal catheter
- Automated measurement collection on LCR Meter for capacitive coupling test method that reduced 99 manual data entry points to one data file through a python script

Dell Technologies — Austin, TX
Mechanical Engineer II | Dell Commercial Mobile Latitude — *April 2023 – June 2025*
- Delivered $500M in cost avoidance by redesigning the LCD assembly, driven by adhesive performance evaluation, mechanical stress analysis, and fatigue testing to develop internal design specifications
- Conducted ANSYS simulations comparing injection-molded hinge designs to identify geometry improvements that reduce stress concentrations and guide future design standards
- Reduced weight of a high-volume platform via mechanical redesign in CAD and material selection, decreasing weight of single platform by 10% while maintaining stiffness and reliability targets
- Reviewed 8 bills of materials with 200+ parts each and drove design changes to reduce costs by 12M USD
- Diagnosed eDP cable design inconsistencies across platforms and partnered with design, electrical, and manufacturing teams to simplify configurations, achieving a 15% reduction in part complexity

Dell Technologies — Austin, TX
Mechanical Engineer I | Dell Commercial Mobile Rugged — *June 2021 – April 2023*
- Developed and filed patent for electromechanical intrusion-prevention design integrating precision sensor and secure mechanical engagement to protect customer data
- Built I/O port housing in CREO and analyzed in ANSYS, reducing testing time by one week
- Created a Power BI dashboard comparing mobile platforms and competitors, enabling faster executive decision-making
- Led a cross-functional team to design and pitch a growth framework aimed at scaling a $240M security business to $2B

Dell Technologies — Austin, TX
Mechanical Engineer Intern | Dell Desktop Precision — *May 2020 – August 2020*
- Prototyped housing for removable SSD cards in Creo for Precision tower quadrupling memory capacity
- Collaborated with engineering team to design module that increases capacity of Precision towers by 8 TB

Ethicon — Cincinnati, OH
R&D Intern | Laparoscopic Surgical Products — *January 2019 – May 2019*
- Designed and built a test fixture for laparoscopic surgery simulation enabling realistic use of laparoscopic devices

Projects

Insulator Defect Identification | *Deep Learning – YOLOv8, Ultralytics, PyTorch*
- Wrote and trained YOLOv8 object-detection model to identify defective power-line insulators in aerial drone imagery, achieving 99.5% mAP@50 on string detection and 92.8% mAP@50 on shell-defect classification

Bio-Inspired Robot for Multi-modal Locomotion
- Performed motor torque analysis for a 7-module bio-inspired snake robot to inform actuator selection
- Executed physical launch testing and statistical analysis (Anderson-Darling normality tests, two-sample t-tests) across 20 trials, demonstrating a statistically significant 8.7% increase in glide distance with undulation at 1.8 m/s ($p = 0.010$)

Skills

CAD, Simulation & Hardware: SolidWorks, Creo, Finite Element Analysis, Gazebo, Mujoco, Arduino, LabVIEW, rapid prototyping
Programming: Python, MATLAB, ROS 2; Machine Learning: PyTorch, OpenCV, Ultralytics, Hugging Face Transformers
Data Analytics: applied statistics, Power BI, reliability testing, JIRA, Agile